What we would improve with more time

Project reflection / Group 4 / One-page technical draft

The strongest result

The useful outcome is a repeatable release decision with evidence. Ridge beat the larger baseline models on validation data and achieved a final historical test MAE of 5.775 lakhs. Two deliberately invalid update scenarios preserved the existing model and prediction, and a second valid release was rolled back successfully. This gives the accuracy result operational context.

What the batches revealed

Batch 2 shifts toward smaller properties without crossing the configured MAE alert. Batch 3 changes both the property mix and prediction error: MAE rises to 13.113 lakhs. This illustrates why valid ranges and stable missingness do not imply stable model behavior, and why input drift must be separated from labeled performance drift. Thresholds were designed after exploratory inspection, so these batches are a demonstration rather than a blind validation of the monitoring policy.

The next engineering priorities

1. Verify the release environment. Run the clean install, actual Docker deployment, GitHub checks and image rollback on an unrestricted host. Benchmark real HTTP traffic with cold starts, concurrent requests and sustained load. The current in-process timing does not establish production capacity.

2. Strengthen transactional deployment. Replace the local pointer/alias coordination with a durable promotion journal and crash recovery; add health-checked blue/green serving, authenticated operations and a shared locking strategy. Keep the current explicit gates and model provenance.

3. Improve monitoring calibration. Calibrate effect thresholds on repeated reference samples, study false-alert rates and seasonal patterns, and report uncertainty. Add subgroup errors and minimum sample sizes. Connect alert delivery to an acknowledged incident workflow.

4. Improve data and model governance. Obtain dated listings for time-based evaluation, document data origin and labeling delay, track approved retraining datasets, and use an untouched prospective evaluation window. Extend reproducibility across operating systems before promising tight numerical tolerance.

Scope and learning

A lightweight local workflow is easy to inspect but does not provide distributed scheduling, high availability or multi-user security. The final delivery must distinguish real local evidence from prepared infrastructure and missing account-dependent evidence. Both team members should review this draft, explain the tradeoffs in their own words, make genuine contributions, and practice a live modification before presenting.